Commercial Fruit Crop Production and Post-Harvest Technology



Class 11 (Technical)

Unit – 3

Cultivation of Fruit Crops



Citrus (Mandarin Orange, Sweet Orange, Lime, Lemon)







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Introduction

❖Botanical Name:



	Mandarin Orange (Suntala)	Sweet Orange (Junar or Mausam)	Lime (Kagati)	Lemon (Nimbuwa)
Scientific Name:	*Citrus reticulata*	*Citrus sinensis*	*Citrus aurantifolia*	*Citrus limon*
Family:	Rutaceae	Rutaceae	Rutaceae	Rutaceae
Origin:	Southeast Asia	Southern China	India (East)	India or Myanmar

❖ It is a perennial either a tree or a Shrub

Introduction

Importance:

- Fruits are used as squash, juice, marmalade, fresh fruit, fruit dessert, lemonade etc.
- They are also used to make cosmetic fragrances.
- They are rich in Vitamin C.
- They are rich in Citric acid.



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Area, Production & Productivity

- Major citrus producing districts are Dhankuta, Panchthar, Dhading, Kavre, Gorkha, Dailekh & Sallyan.
- Mandarin covers about 62%, Junar 22%, Lime 12%, Lemon 3% and other 1% of the total citrus production.





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Soil & Climate

- ❑ It can be cultivated in wide range of soil but deep, loose with loam and sandy loam is ideal for commercial cultivation.
- ❑ The soil should have high OM and have proper drainage facility.
- ❑ pH should be 5.0 - 6.5

- ❖ Mandarins perform better at higher altitudes, sweet oranges at lower hills while lime & lemon do best on low hills to terai.

	Mandarins Orange	Sweet Orange	Lime	Lemon
Altitude:	800-1400 msl	800-1300 msl	500-1000 msl	500-1000 msl
Temperature:	10-30°C	10-30°C	10-30°C	10-30°C
Rainfall:	100-200 cm/yr	150-200 cm/yr	100-150cm/yr	100-150cm/yr

Propagation

- Both seed and vegetative methods are used.
- Budding & Grafting are commercially practiced asexual methods.
- Among several rootstocks used, Trifoliate is a good rootstock. It is resistant to root rot, nematodes & cold stress. It tolerates the Tristeza virus as well.





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Cultivars

Mandarine:

Pokhara Local,
Murcott,
Chunakawari
Dhankuta Local
Nagpuri

Sweet Orange:

Junar,
Mausambi,
Valencia,
Malta Blood Red
Satgudi
Novel orange
Pineapple

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Cultivars

Lime:

Tehrathum local,
Kagzi lime,
Mexican lime,
Persian lime,
Bears Seedless
Banaras local
Rampur-01 & 10
Rampur-101

Lemon:

Sunkagati 1,
Nepali amilo,
Suraka round,
Libson lemon,
Sunkagati 2
Assam amilo
Eureka
Lucknow seedless

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Planting

- In a well prepared land sapling should be planted in a pit size of 70*70*70 cm.
- The best time for citrus planting is before onset of monsoon or end of monsoon.
- June-July to September (Mandarine)
- The spacing should be maintained about 4-5 m, but it depend on variety.
- Generally square or contour system is adopting.








Lime Nutrition

VITAMINS

- **C** — 29.1 mg (32% DV) Ascorbic acid
- **B_5** — 0.217 mg (4% DV) Pantothenic acid
- **B_6** — 0.043 mg (3% DV) Pyridoxine
- **B_1** — 0.03 mg (3% DV) Thiamin
- **B_9** — 8 µg (2% DV) Folate
- **B_2** — 0.02 mg (2% DV) Riboflavin

Less than 2% DV vitamins: Choline, E, K, B_3, & A

CALORIES

30 kcal



Serving Size: 100 g

MINERALS

- **Cu** — 0.065 mg (7% DV) Copper
- **Ca** — 33 mg (3% DV) Calcium
- **Fe** — 0.6 mg (3% DV) Iron
- **K** — 102 mg (2% DV) Potassium

Less than 2% DV minerals: Phosphorus, Magnesium, Zinc, Selenium, & Manganese

WATER	PROTEIN	CARBOHYDRATE	TOTAL FATS	DIETARY FIBER	SUGARS
88.26 g	0.7 g (1% DV)	10.54 g (4% DV)	0.2 g (0% DV)	2.8 g (11% DV)	1.69 g (3% DV)

Sources: *USDA National Nutrient Database*
Average Daily Values reference: *NHI Dietary Supplement Label Data Base*

HerbaZest®

Lemon Nutrition

VITAMINS

- **C** — 30.7 mg (34% DV) Ascorbic acid
- **B_6** — 0.046 mg (3% DV) Pyridoxine
- **B_9** — 6 µg (2% DV) Folate
- **B_5** — 0.11 mg (2% DV) Pantothenic acid
- **B_1** — 0.023 mg (2% DV) Thiamin

Less than 2% DV vitamins: B_2, Choline & E

CALORIES

17 kcal



Serving Size: 1 Lemon= 58 g

MINERALS

- **K** — 80 mg (2% DV) Potassium
- **Fe** — 0.35 mg (2% DV) Iron
- **Cu** — 0.021 mg (2% DV) Copper

Less than 2% DV minerals: Calcium, Phosphorus, Magnesium, & Manganese

WATER	PROTEIN	CARBOHYDRATES	DIETARY FIBER	SUGARS
51.61 g	0.64 g (1% DV)	5.41 g (2% DV)	1.6 g (6% DV)	1.45 g (3% DV)

Sources: USDA National Nutrient Database
Average Daily Values reference: NHI Dietary Supplement Label Data Base

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Orange Nutrition

VITAMINS

- 69.7 mg (77% DV) Ascorbic acid — C
- 39 µg (10% DV) Folate — B_9
- 0.114 mg (10% DV) Thiamin — B_1
- 0.328 mg (7% DV) Pantothenic acid — B_5
- 0.079 mg (5% DV) Pyridoxine — B_6
- 0.052 mg (4% DV) Riboflavin — B_2
- 14 µg (2% DV) Vitamin A, RAE — A
- 11 mg (2% DV) Choline
- 0.369 mg (2% DV) Niacin — B_3
- 0.24 mg (2% DV) Alpha-Tocopherol — E

CALORIES

62 kcal



MINERALS

- Cu 0.059 mg (7% DV) Copper
- K 237 mg (5% DV) Potassium
- Ca 52 mg (4% DV) Calcium
- Mg 13 mg (3% DV) Magnesium

Less than 2% DV minerals:
Phosphorus, Iron, Selenium, Zinc, & Manganese

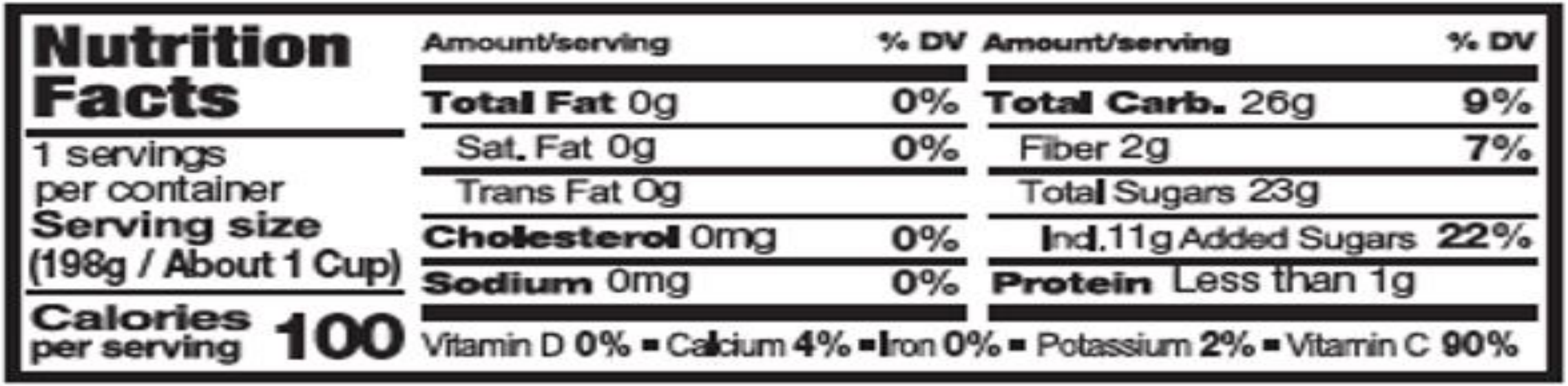
Serving Size: 1 fruit (2-5/8" dia)= 131 g

WATER	PROTEIN	CARBOHYDRATES	FATS	DIETARY FIBER	SUGARS
113.64 g	1.23 g (2% DV)	15.39 g (5% DV)	0.16 g (0% DV)	3.1 g (12% DV)	12.25 g (25% DV)

Sources: USDA National Nutrient Database for Standard Reference
Average Daily Values reference: NHI Dietary Supplement Label Data Base

HerbaZest

MANDARIN ORANGE



Nutrition Facts

1 servings per container
Serving size
(198g / About 1 Cup)

Calories per serving **100**

Amount/serving	% DV	Amount/serving	% DV
Total Fat 0g	0%	**Total Carb.** 26g	9%
Sat. Fat 0g	0%	Fiber 2g	7%
Trans Fat 0g		Total Sugars 23g	
Cholesterol 0mg	0%	Incl. 11g Added Sugars	22%
Sodium 0mg	0%	**Protein** Less than 1g	

Vitamin D **0%** • Calcium **4%** • Iron **0%** • Potassium **2%** • Vitamin C **90%**

A medium mandarin orange (about 88 grams) provides approximately 47 calories, 12 grams of carbohydrates, 1 gram of protein, and 0.3 grams of fat. It's also a good source of vitamin C, with 26% of the Daily Value, as well as fiber, magnesium, potassium, and copper.

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Training & Pruning

- The diseased, dried, unwanted and overlapped branches are removed. The suckers are also removed.
- Light pruning should be done to improve productivity and reduce pest problems.
- The main trunk should be cut at the height of 1m which helps to develop lateral branches. 3-5 lateral branches are kept.





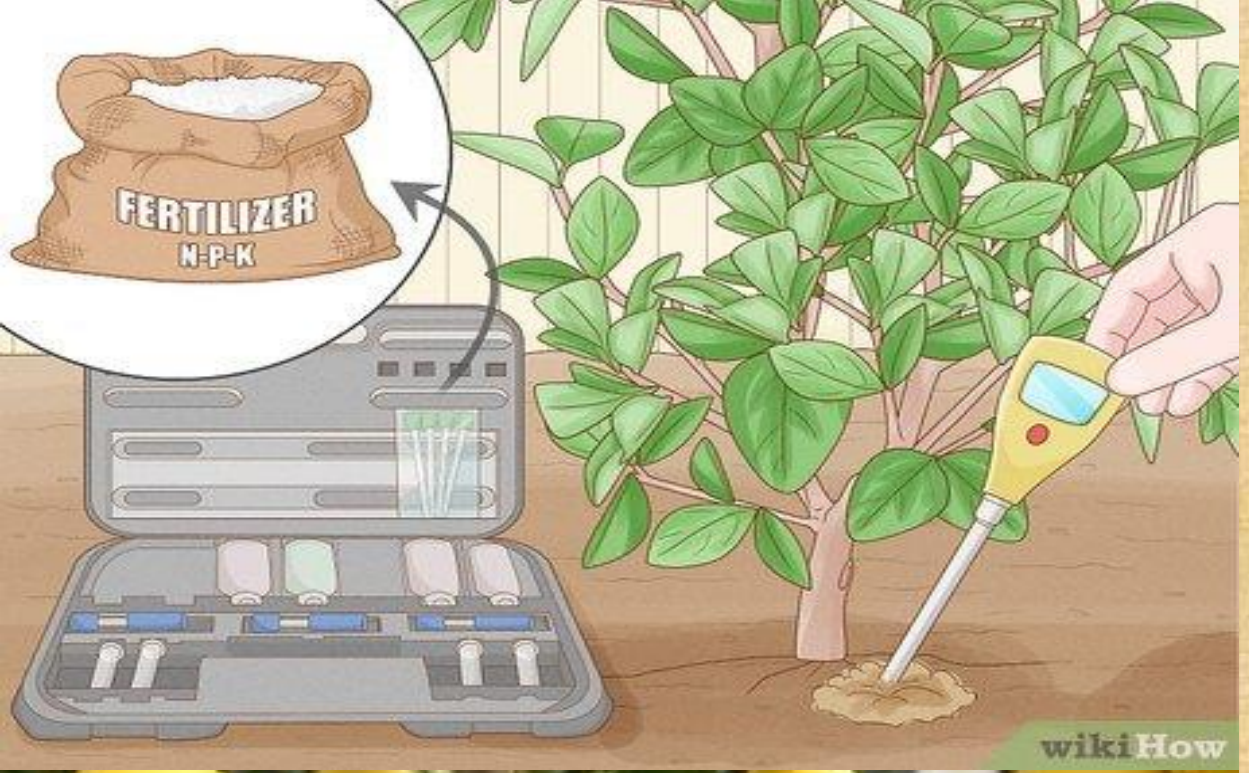

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Cultural Operation

A. Manure & Fertilizer Application

- In general, fruit bearing tree should be given 50-80 kg compost, 800g N, 500g P & 800g K/year/tree.
- Half dose (Magh - Falgun) remaining dose (Jesth - Asar)


FERTILIZER
N-P-K
wikiHow


Once every 1-2 months
FERTILIZER
wikiHow




wikiHow

Cultural Operation

B. Intercropping

- Shallow rooted, non-exhaustive vegetables, legumes & shade loving crops can be taken as intercrops. But, intercrop is not desirable in older plantations when spaces with adequate light are diminished.
- When planting distance is more than 4m x 4m then intercrops can be grown for the first two to three years.

- Proper weeding should be done whenever necessary mainly at the time of irrigation.

Cultural Operation

C. Irrigation

- Irrigation depend upon soil type and availability of water.
- Irrigation is more essential during plant growth, flowering and fruiting stage.
- Less soil moisture may cause flower and fruit drop .
- Generally, irrigation is done at 15-30 days interval depending upon the soil moisture.
- Heavy irrigation may cause less flowering & root rotting.





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Disease Management

1. Die-Back **C/O**: Fungus **S.N:** *Colletotrichum gloeosporioides*

Symptoms:

- The leaves become yellow in initial stage and turn into dark colour and start to fall with beginning of summer.
- Die-back of twig starts leading to dry and finally the plant dies.

Management:

- Remove dead twigs or affected part from the plant.
- Spray Carbendazim @ 1g /l of water.
- Bordeaux mixture 2:2:250 in April, July-August and September spray.





Disease Management

2. Phytophthora rot (Root rot, Crown rot, gummosis)

C/O: Fungus **S.N:** *Phytophthora parasitica* *P. citrophthora*

Symptoms:

- Dark discoloration of bark is developed, water soaked spots developed which cause stem rot, rootlet rot and finally defoliation.
- In advance stage bark cracking and oozing of gums takes place.

Management:

- Avoid water logging condition at the base of the plant.
- Scrap the affected part.


J.H. Graham, CREC



Disease Management

3. Citrus Canker **C/O:** Bacteria **S.N:** Xanthomonas citri

Symptoms:

- It is a serious disease in citrus fruits.
- It affects the quality of fruit and reduces market value.
- Water soaked, round spots are seen on the leaves, stems & in fruits.
- The spots increases slowly and turn into brownish on the leaves, stems and fruits.

Management:

- Remove and destroy the infected part of the plant.
- Spray Bordeaux mixture @ 1% on cut portion.
- Streptocycline @ 1g/l of water initially.




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Disease Management

4. Greening **C/O:** Bacteria **S.N.:** *Libribacter asiaticus*

Symptoms:

- Initially vein and veinlet turns yellow, then interveinal chlorosis is observed.
- Leaves become, small, pair in mass and bend downward of seasonal flowering. Die-back symptom is another major symptom of this disease.
- Division of cloves are not in equal proportion and seeds are not viable.
- It is transmitted by citrus phyla.

Management:

- The infected parts are destroyed. Spray Diemethoade @ 2-3ml/l of water to control the vector citrus phyla.
- Apply Bordo pest in cut portion. Spry Streptocylin @1g/l of water after pruning.





Disease Management

5. Tristeza Virus

Symptoms:

- It is a serious disease of citrus family.
- It cannot be controlled easily.
- Initial symptoms are seen like development of small leaf like green in colour, small fruit with sour test.
- At later stage, the plant wilts and dies.

Management:

- Destroy infected plant part.
- Spray Dimethoate 30% EC @ 2 ml/l of water to control sucking pest.


a
b
c
d

Physiological Disorder

1. Fruit drop:

- Fluctuating temperature, low atmospheric humidity, imbalance of soil moisture, lack of proper nutrition, hormonal imbalance, incidence of insect pests and diseases are some factors causing fruit drop.

2. Granulation:

- Fruit become comparatively hard and become somewhat enlarged.
- Spraying of lime reduces the extent of granulation. Similarly applications of 2,4-D (12ppm), zinc & copper reduces the incidences of granulation considerably.





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Physiological Disorder

3. Decline:

- After fruitful production from about 5 years, citrus orchards start bearing little crop and become uneconomical.
- Good cultural practices, improvement in soil fertility and drainage, control of insect pests, nematodes and diseases may be useful to minimize its incidence.


6



Pest Management

1. **Aphid** (Toxoptera citricida, Myzus percicae, Aphis gossipi)

Symptoms:

- The adult and nymph suck the sap from tender parts of the plant, leaves, flowers and fruits.
- It causes the stunted growth of plant and reduction in yield.

Management:

- Adopt clean cultivation.
- Spray Dimethoate 30% EC @ 1-2 ml/l or Imidachloprid 17.8% SC 0.3 ml per litre of water.
- Use lady bird beetle to control the population of aphid.







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Pest Management

2. Scale insect *(Aonidiella auriantii, A. citrina)*

Symptoms:

- They suck the sap from tender parts of the plant, leaves, flower and fruit.
- The affected part gets damage which result premature fruit drop and defoliation.

Management:

- Adopt clean cultivation.
- Spray the Dimethoate 30%EC @ 1-2ml/l of water.





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Pest Management

3. Citrus Psylla (Diaphorina citri)

Symptoms:

- They suck the sap from the tender parts of the plant, leaves, flowers and fruits.
- They create honey dew on the leaves in which shooty mould is developed.

Management:

- Establish nursery about 1300masl.
- Destroy the secondary host i.e. Kamini flower of this insect.
- Spray Dimethoate 30%EC @ 2-3ml/l of water.





Pest Management

4. Lemon butterfly *(Papilio demoleus)*

Symptoms:

- Female lays eggs on the fruit.
- After hatching larva feed on the pulp of the fruit causing rooting and dropping of fruits.

Management:

- Collect and destroy the affected and falling fruits.
- Destroy the larva and eggs of the insect.







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Pest Management

5. Leaf Miner *(Phyllocnistis citrella)*

Symptoms:

- Small larva bore in the tissue of the leaves and make zig-zag tunnel.
- Feed on the chlorophyll internally due to which the tunnel looks silver color.

Management:

- Spray tobacco based pesticide. e.g. Soak 250 tobacco in 400 liter of water and mix with 30g surf. Keep it whole night and filter it. Now add one part of this solution in a part of water and spray at 10 days interval. Apply 0.06% solution of Thiamethoxam 25 WG





Fruiting

- Flower takes place in the month of Jestha - Ashar.
- The fruitlets develop in the month of Bhadra – Ashoj.
- Low humidity favors color development, whereas plants not having adequate sunlight produce low yields of poor quality fruits.
- Some citrus tree tend to have a bearing habit with alternating high and low yields.





Harvesting & Yield

- Fruits should be harvested when they attain full size, develop attractive color with optimum sugar : acid ratio.
- Harvesting should be done carefully without causing damage to the fruits.
- Slight bruising or injury to the rind leads to quick decay of fruits.
- Harvesting can be done by shaking the branches with a pole and also by hand using a ladder.

- Generally, citrus tree yields about 10-12 tones/ha. But it vary as per varieties, care and management.

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Post Harvest Handling

Grading and Packaging:

- Generally, citrus are graded according to their size and appearance.
- Fruits are usually packed in wooden boxes for distant markets, while for local marketing baskets of split bamboo and mulberry are used.
- Chopped straw and dry grasses are mostly used for padding.
- The fruits should be cleaned and polished lightly with a piece of cloth, before wrapping them in tissue paper or newspaper.
- Use of corrugated fiber board cartons in place of wooden boxes is highly beneficial.

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Post Harvest Handling

Transport and storage:

- Citrus are generally transported by road, without refrigeration which often leads to heavy losses due to decay and fungal infection.
- Use of shrink films for moisture loss control can be useful.
- Storage life of citrus is influenced by many pre and post harvest factors.
- Green or fully ripe citrus fruit can be stored successfully at 8-10°C with 85-90% relative humidity without loss of fruit quality.





Marketing

- Citrus has National & International demands, so that we can supply those market.
- To get higher price fruits must be graded properly based on size, variety trueness etc.

Thank You For Your Attention……

Any Question ???

See You In Next Class !!

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